

WHERE DO CLIMATE-AFFECTED HOUSEHOLDS GO?

Revealed District Choice in Bangladesh

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ABSTRACT

Climate-mobility research usually asks whether an environmental shock changes the probability of moving. This paper begins after a move has occurred and asks which destination is chosen. I construct revealed district-choice events from two independent Bangladesh surveys and compare every observed destination with the same 64-district candidate set. The strongest transport test uses 1,857 current migrants in the nationally representative rural Bangladesh Integrated Household Survey (BIHS), spanning all 64 origin districts. A pre-specified GIS model adds historical flood exposure, built surface, travel time to cities, and cropland to a fitted distance-population gravity model. GIS lowers held-out log loss by 0.108 in household-grouped validation, by 0.101 when each origin district is excluded from training, and by 0.094 when a model trained on 2015 predicts 2018-19 moves. Climate-specific tests preserve the same outcome and feature set: the gain is 0.098 for 123 BIHS relocations attributed to erosion-related land loss and 0.108 for 184 shock-linked relocations in the Bangladesh Environmental Mobility Panel (BEMP). This is distinct from prior BEMP research on whether shocks change migration likelihood, type, or distance: the sample here is conditional on observed movement, and the outcome is the destination district. In the smaller climate samples, full-choice performance does not transfer cleanly to unseen origins, although the interdistrict advantage remains positive. District characteristics therefore carry portable information about destination ranking, but local context remains important for whether a move stays inside the origin district. The analysis is predictive, not causal.

Keywords: climate mobility; destination choice; Bangladesh; riverbank erosion; GIS; out-of-sample prediction

1. Introduction

In 2010, an anonymized household head in the BIHS reported leaving Faridpur for Manikganj after river erosion caused the loss of land or homestead land. The move appears in a survey row, stripped of a name and most of the household's story. It still poses a concrete question. Why Manikganj rather than one of Bangladesh's other districts?

Most climate-mobility research begins at the origin. It estimates whether floods, erosion, drought, or crop loss change migration rates. That work has corrected the simple idea that environmental damage always produces long-distance displacement. In rural Bangladesh, disasters can increase local movement, reduce movement by destroying the resources needed to leave, or have little measurable effect on mobility (Gray and Mueller 2012). Environmental pressures also combine with economic, social, demographic, and political conditions rather than operating alone (Black et al. 2011).

The destination receives less attention. Findlay (2011) argued that climate-mobility research should study where observed movers go instead of concentrating only on projected counts. That shift matters for planning. A household's origin identifies exposure and loss. Its destination determines the labor market, housing conditions, agricultural base, infrastructure, social ties, and future hazards the household encounters.

This paper treats an observed move as a choice among candidate districts. It asks whether a small set of mapped destination attributes predicts that choice better than two familiar spatial baselines. The gravity model rewards large, nearby destinations. The radiation model also accounts for population between origin and destination (Simini et al. 2012). The GIS model retains distance and population, then adds flood history, settlement intensity, urban accessibility, and cropland.

The study is not a re-estimation of the existing BEMP migration-incidence result. Freihardt (2025) asks whether environmental shocks change the probability, type, and distance of migration among climate-vulnerable households. I condition on a recorded move and make the chosen destination district the outcome. BEMP supplies the strongest shock-to-move timing, but its seven observed climate-sample origins cannot support a national transport claim. For that reason, the empirical argument begins with BIHS: an independent survey with all 64 origin districts, a later-wave test, and an erosion-specific replication.

The national BIHS result is the main portability test. GIS improves destination prediction among 1,857 migrants, survives exclusion of each test origin from training, and transfers from 2015 to 2018-19. The two climate-specific samples then ask whether the same pre-specified destination model holds in a narrower substantive setting. It does under household-grouped validation. The harder origin-held-out results expose a boundary: predicting whether a move remains inside an unseen origin district is less portable than ranking destinations after an interdistrict move is known.

The contribution is deliberately narrow. Four district characteristics add repeatable predictive information beyond distance and population. Individual coefficients do not describe causal preferences. Districts also hide movement within villages and neighborhoods. Those limits define what the public survey geography can support.

2. Destination choice, spatial interaction, and environmental loss

2.1 Destination choice as spatial utility

Migration can be understood as an investment whose expected returns are weighed against monetary, informational, social, and psychological costs (Sjaastad 1962). That decision is spatial because neither returns nor costs are constant across destinations. In a random-utility formulation, a mover assigns each candidate place a systematic value and an unobserved household-place component, then selects the alternative with the highest utility (McFadden 1974). Conditional-logit migration models make this comparison explicit by placing the attributes of chosen and unchosen destinations in the same choice set (Davies, Greenwood, and Li 2001).

Gravity models capture two durable parts of this structure. Destination population approximates opportunity mass, while distance absorbs travel cost, information loss, separation from social ties, and other frictions. Yet distance and size do not exhaust spatial choice. Competing-destinations theory argues that the attractiveness of one place depends on the alternatives around it, not only on its bilateral relation to the origin (Fotheringham 1983). The radiation model formalizes a related intuition through the population opportunities between origin and destination (Simini et al. 2012). These approaches motivate demanding baselines: GIS should add value only if destination structure contains information beyond proximity, scale, and intervening opportunity.

2.2 Migration systems under environmental stress

Migration systems are sustained by feedback among origins, destinations, earlier movers, information, and institutions (Mabogunje 1970). Environmental loss enters that system as a constraint and an adjustment pressure rather than as a mechanical command to move (Wolpert 1966; Black et al. 2011). Erosion can remove land and housing. Flooding can damage crops, assets, transport, and local work. The same shock can make departure more urgent while destroying the resources needed to travel, rent, or rebuild.

Destination choice therefore follows a constrained comparison. A household may value wage access, housing, cultivable land, social support, or safety from a repeated hazard, but it can choose only among places it knows and can reach. The analytical set of 64 districts is an evaluation universe, not a claim that every respondent consciously considered every district. It provides a consistent set of chosen and unchosen alternatives against which models can be tested.

The four GIS variables are common proxies for parts of this destination system. Built surface is not a wage; cropland is not a job offer; travel time is not a household's actual trip; historical flood exposure is not perceived safety. They describe candidate districts where direct, pre-move measures of jobs, rents, land access, and social ties are unavailable. Their coefficients should therefore be read as predictive weights, not estimates of household preferences or causal effects.

2.3 What this study adds to the BEMP evidence

Freihardt (2025) uses BEMP to estimate how environmental shocks affect migration likelihood, type, and distance. Related work describes short-distance mobility, land access, and social ties among Jamuna River households (Freihardt 2026). Those are origin-side and migration-incidence questions: did a household move, what kind of move occurred, and how far did it go?

This paper asks a different conditional question. After a move is observed, which district receives it? The unit of analysis is an event-candidate pair, the outcome is the reported destination district, and every observed destination is evaluated against unchosen districts. BEMP is used for its unusually strong sequence of recorded shocks and later relocations. BIHS is not a supplementary robustness check; it is an independent replication with broader origin support and the main test of whether destination utility travels across places and time. No result here estimates the causal effect of an environmental shock on migration incidence or the causal effect of a destination attribute on choice.

2.4 Predictions

The theory yields three tests. First, destination GIS characteristics should lower held-out log loss relative to fitted gravity. Second, if they capture portable destination utility rather than origin-specific regularities, that advantage should survive later-wave prediction and the exclusion of each test origin from training. Third, portability should be weaker for the full 64-district task than for the interdistrict task. The full task combines whether a move stays inside the origin with where it goes; the latter conditions on crossing the district boundary.

3. Data

3.1 Bangladesh Integrated Household Survey

BIHS is a nationally representative rural household panel administered in 2011-12, 2015, and 2018-19 by the International Food Policy Research Institute. It samples all administrative divisions and contains detailed modules on households, agriculture, shocks, migration, and remittances (IFPRI 2011, 2016, 2020).

I construct two BIHS designs. The 2015 household-relocation module records the previous district, current district, year, and reason for the household head's last change of residence. The main climate replication contains 123 moves explicitly attributed to losing land or homestead land through river erosion. These events cover 29 origin districts, and 71 cross a district boundary. A broader comparison uses all 526 district-resolved household relocations in the module.

The second design uses current migrants reported in the 2015 and 2018-19 migration modules. The conservative event ledger keeps 1,857 interval-specific domestic migrants from 1,404 households. Each person was away for at least six months and outside the origin upazila. The sample covers all 64 origin districts and 63 observed destination districts; 1,208 events cross a district boundary. It records a stock of current individual migrants, not complete migration spells or whole-household displacement.

BIHS supplies broader geographic coverage and an independent erosion measure. Its environmental timing is weaker than BEMP's. Some household relocations are retrospective, and the current-migrant sample should not be described as a climate-migration sample.

3.2 Bangladesh Environmental Mobility Panel

BEMP followed 1,691 households and 2,170 panel respondents from 2021 through 2024 in communities selected along the eastern bank of the Jamuna River. The public release contains four annual in-person rounds and ten shorter phone rounds (Freihardt, Rudolph, and Koubi 2026). Migrants were followed across survey waves.

The event ledger uses new household-destination observations and reconstructs the latest valid pre-move district. A shock-linked event requires a recorded home flood or riverbank-erosion shock before the move interval. Public data identify origin and destination districts, not exact household coordinates. The primary BEMP sample contains 184 whole- or partial-household relocations from 137 households. Seventy-one cross a district boundary; 113 remain within the origin district.

BEMP offers the strongest temporal connection between environmental exposure and a later move. Its geographic support is narrow. The climate sample comes from seven observed origin districts and concentrates heavily in a small set of destinations. National candidate coverage cannot turn those origins into a national sample.

3.3 Choice sets and administrative geography

Every event receives the same national district universe. The full task includes all 64 districts, including the origin district. It jointly evaluates staying within the origin district and moving elsewhere. The interdistrict task excludes the origin and evaluates only observed cross-district moves against 63 candidates.

District names from BEMP, BIHS, and GIS sources are reconciled to a frozen Bangladesh Bureau of Statistics crosswalk. Population comes from the 2022 Population and Housing Census (Bangladesh Bureau of Statistics 2023). District polygons follow the public BBS administrative boundary release. The same crosswalk and candidate ordering are used throughout.

Table 1. Analysis samples

Dataset and sample	Unit	Events	Cross-district events	Observed origins	Timing strength
BIHS erosion relocations	Household-head residential move	123	71	29	Retrospective reason: erosion-related land loss
BIHS all household relocations	Household-head residential move	526	236	61	Retrospective move history
BIHS interval migrants	Current individual migrant	1,857	1,208	64	Interval-specific current migrant stock
BEMP shock-linked relocations	Whole/partial household move	184	71	7	Lagged recorded flood or erosion before move interval

4. Destination measures

The gravity benchmark contains log destination population and log effective distance. Great-circle distance between district centers is adjusted for district size. This gives the within-district alternative a finite positive distance rather than zero.

The GIS model adds four destination measures selected before the BIHS outcomes were examined:

1. **Historical flood exposure:** the share of valid district land flooded in at least one selected Global Flood Database event from 2000 through 2018. The source uses 250-meter MODIS observations of 913 large floods (Tellman et al. 2021).

2. **Built surface:** built square meters divided by valid district land area from the 2020 epoch of GHS-BUILT-S R2023A (Schiavina et al. 2023).
3. **Urban accessibility:** the area-weighted median travel time to the nearest city of at least 50,000 people using the 2015 global accessibility surface (Weiss et al. 2018).
4. **Cropland:** the share of valid district land classified as cropland in ESA WorldCover 2020 at 10-meter resolution (Zanaga et al. 2021).

All four variables cover every district. Raster extraction uses land masks, boundary-cell fractions, and area-aware calculations. Each training fold supplies its own means and standard deviations for GIS standardization. The test fold never contributes to scaling.

The timing contract warrants care. The GIS layers provide a common district description, not event-year causal exposures. BEMP move intervals reach 2021-24, BIHS migration records extend further back, and 2022 population postdates some moves. The variables therefore enter a prediction model. They should not be read as treatments occurring before every household decision.

5. Empirical design

For mover or household event i , origin o , and candidate destination j , the gravity utility is

$$U_{ioj}^G = \beta_1 \log(Pop_j) + \beta_2 \log(Distance_{oj}).$$

The GIS utility is

$$U_{ioj}^{GIS} = U_{ioj}^G + \gamma' Z_j,$$

where Z_j contains the four standardized GIS attributes. Candidate probabilities follow the conditional-logit softmax over the relevant choice set. The primary GIS estimator applies a ridge penalty to the four GIS coefficients while leaving the gravity coefficients unpenalized. An inner household-grouped validation loop selects the penalty from 0.001, 0.01, 0.1, 1, 10, and 100. A low-dimensional unpenalized fit remains a diagnostic.

The fitted gravity model is the primary comparator. A uniform model and an adapted radiation model provide secondary checks. No destination fixed effects, observed destination frequencies, move reasons, or post-move network reports enter candidate utility.

5.1 Evaluation

The main validation uses five household-grouped folds. All events from the same household remain together, and every transformation is estimated inside the training data. The primary metric is event log loss, the negative log probability assigned to the chosen district. I report GIS gain as gravity log loss minus GIS log loss. A positive value favors GIS. Rank and top-choice accuracy serve as secondary measures.

Paired uncertainty comes from 5,000 household-cluster bootstrap samples of held-out event losses. Because the two models predict the same events, the bootstrap resamples their within-event loss difference rather than treating model results as independent.

Two tests raise the difficulty. The temporal test estimates the national BIHS migrant model on 2015 events and evaluates it on 2018-19 events. Leave-one-origin-out validation trains without any event from the test

origin district. The latter asks whether a destination utility learned elsewhere can score movers from a place absent from training.

The full task and interdistrict task answer different questions. Full-choice performance includes the probability of remaining inside the origin district. Interdistrict performance conditions on crossing the boundary and evaluates the ranking of the other 63 districts. The paper reports both rather than treating same-district movement as an ordinary zero-distance trip.

6. Results

6.1 Independent BIHS replication and transport

The broad BIHS migrant sample provides the strongest geographic test because it includes all 64 origin districts. GIS lowers household-grouped held-out log loss from 2.060 to 1.952 across 1,857 events, a gain of 0.108 [0.082, 0.135]. Among the 1,208 cross-district events, the gain is also 0.108 [0.073, 0.145]. These migrants are not labeled as climate migrants; their role is to test whether the destination specification works across Bangladesh rather than only near the Jamuna River.

The result survives two harder changes in the training data. Leave-one-origin-out validation excludes every observation from the test origin district. The full-choice gain remains 0.101 [0.074, 0.128], and the interdistrict gain is 0.107 [0.073, 0.144]. Forty-six of 64 origins have a positive mean full-choice gain. A temporal test trained on 2015 migrants and evaluated on 1,383 migrants in 2018-19 gives a full-choice gain of 0.094; among 842 cross-district events, the gain is 0.112.

Table 2. Transport tests led by the independent BIHS sample

Sample	Test	Choice set	Events	GIS gain	95% interval
BIHS interval migrants	Leave one origin out	Full 64	1,857	0.101	[0.074, 0.128]
BIHS interval migrants	Leave one origin out	Interdistrict 63	1,208	0.107	[0.073, 0.144]
BIHS erosion relocations	Leave one origin out	Full 64	123	-0.016	[-0.086, 0.045]
BIHS erosion relocations	Leave one origin out	Interdistrict 63	71	0.058	[-0.034, 0.140]
BIHS interval migrants	Train 2015, test 2018-19	Full 64	1,383	0.094	Not bootstrapped
BIHS interval migrants	Train 2015, test 2018-19	Interdistrict 63	842	0.112	Not bootstrapped

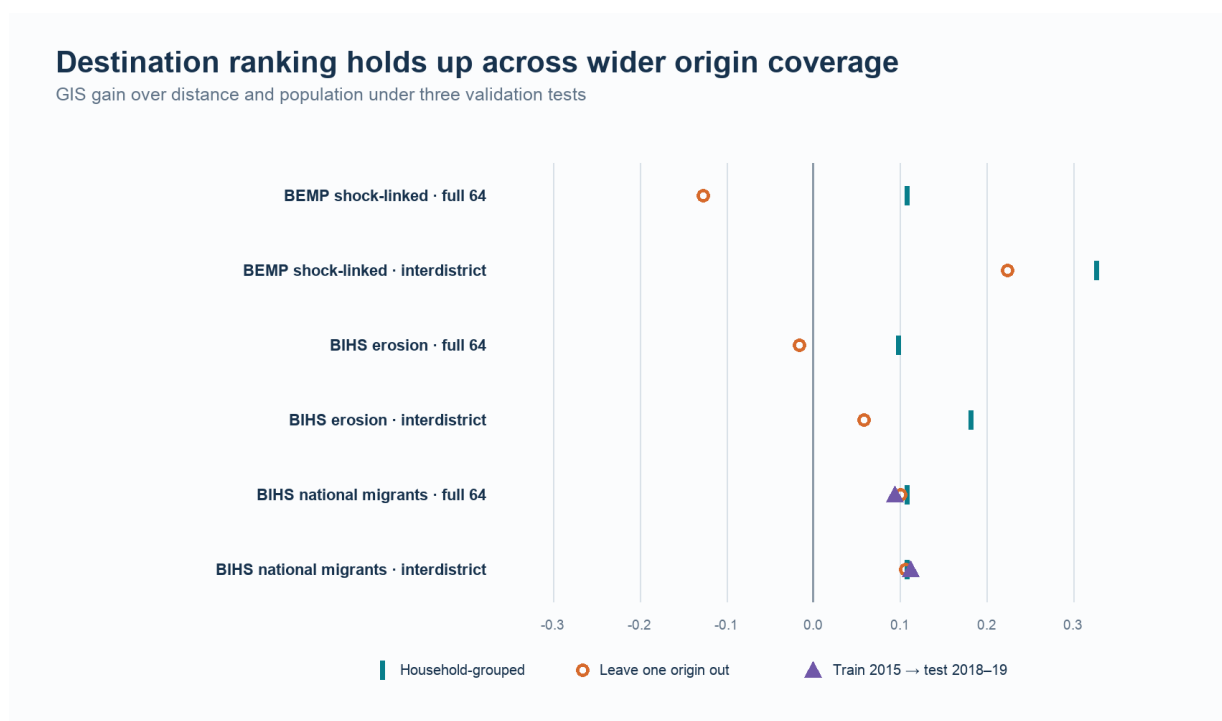


Figure 1. GIS gain under grouped, origin-held-out, and temporal validation.

6.2 Climate-specific destination prediction

The erosion-specific BIHS sample is an independent climate replication with a narrower sample but wider origin support than BEMP. Across 123 household relocations attributed to erosion-related land loss, GIS lowers full-choice log loss from 2.563 to 2.465, a gain of 0.098 [0.028, 0.163]. Among 71 cross-district relocations, the gain is 0.182 [0.076, 0.287].

BEMP supplies the stronger sequence of a recorded flood or erosion shock followed by a later move. Across 184 shock-linked relocations, GIS lowers full-choice log loss from 1.630 to 1.522, a gain of 0.108 [0.023, 0.189]. Its interdistrict gain is 0.327 [−0.001, 0.634] across 71 events. The close full-choice estimates—0.098 in BIHS and 0.108 in BEMP—come from different surveys, migration definitions, and origin supports; their similarity was not imposed.

Table 3. GIS gain over fitted gravity in household-grouped validation

Dataset and sample	Choice set	Events	Gravity log loss	GIS log loss	GIS gain	Household-cluster 95% interval
BIHS interval migrants	Full 64	1,857	2.060	1.952	0.108	[0.082, 0.135]
BIHS interval migrants	Interdistrict 63	1,208	2.021	1.913	0.108	[0.073, 0.145]
BIHS erosion relocations	Full 64	123	2.563	2.465	0.098	[0.028, 0.163]
BIHS erosion relocations	Interdistrict 63	71	3.256	3.074	0.182	[0.076, 0.287]
BIHS all household relocations	Full 64	526	2.206	2.148	0.058	[0.030, 0.088]
BIHS all household relocations	Interdistrict 63	236	3.253	3.138	0.116	[0.056, 0.179]
BEMP shock-linked relocations	Full 64	184	1.630	1.522	0.108	[0.023, 0.189]

Dataset and sample	Choice set	Events	Gravity log loss	GIS log loss	GIS gain	Household-cluster 95% interval
BEMP shock-linked relocations	Interdistrict 63	71	2.484	2.157	0.327	[−0.001, 0.634]

The nested full-choice model treats staying and crossing as separate stages. Its GIS gains over direct gravity are 0.216 for the national migrant sample, 0.129 for BIHS erosion relocations, 0.116 for all BIHS household relocations, and 0.124 for BEMP shock-linked moves. The adapted radiation model performs worse than fitted gravity in the BIHS interdistrict samples. Population between origin and destination does not replace the four destination measures in this application.

The climate-specific origin-held-out tests are less stable. BIHS erosion relocations give a full-choice gain of −0.016 [−0.086, 0.045] and an interdistrict gain of 0.058 [−0.034, 0.140]. BEMP shows the same directional split: −0.127 for full choice and +0.224 for interdistrict choice. BEMP has only seven climate-sample origins, so its origin-held-out estimates cannot support a national claim.

The evidence points to an origin boundary. A model trained elsewhere has difficulty assigning the probability of staying in an unseen erosion-affected district. Once a cross-district move is known, GIS destination ranking remains favorable in direction, although the climate samples are too small to estimate that transfer precisely.

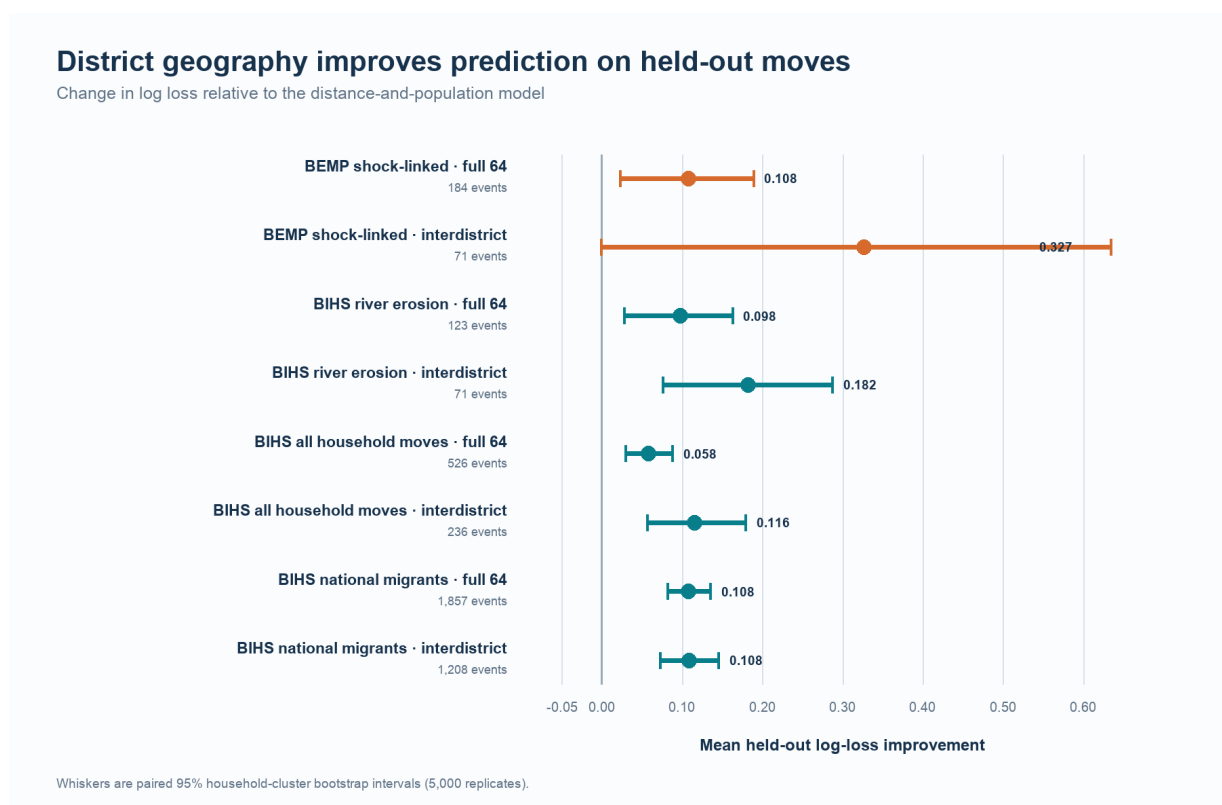


Figure 2. Cross-dataset GIS gain over gravity. Points show mean held-out log-loss improvement; whiskers show paired 95% household-cluster bootstrap intervals.

6.3 One household's observed move

The Faridpur household illustrates what changes when the model sees more than distance and population. Gravity assigned 7.0% probability to Manikganj and ranked it sixth among 64 districts. The GIS model assigned 13.7% and ranked it second. Faridpur remained the model's top candidate, so GIS did not recreate the household's decision perfectly. It made the chosen destination considerably more plausible.

The case does not disclose a name, family history, income, or private deliberation. It cannot show that flood exposure, built surface, accessibility, or cropland caused this household to choose Manikganj. It shows how an observed human journey becomes a transparent prediction test: the model scores every district, and the household's reported destination determines whether the probability assignment improved.

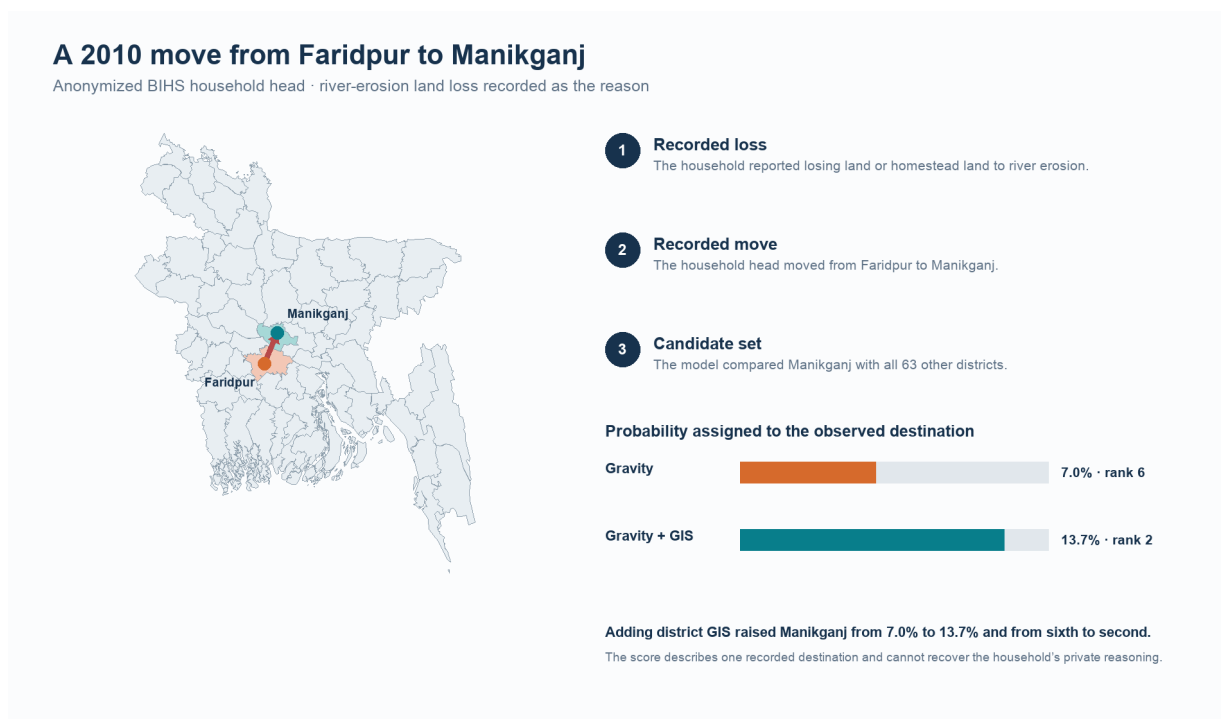


Figure 3. An anonymized BIHS household move from Faridpur to Manikganj after erosion-related land loss.

7. Discussion

The analysis answers the destination question with a qualified yes. Distance and population explain a substantial share of observed movement, especially the prevalence of nearby and same-district moves. Four mapped destination characteristics improve the remaining probability assignment across independent surveys.

The broad BIHS sample carries the strongest transfer result. Its origin coverage reaches all 64 districts, its improvement survives a later wave, and it remains positive when every test origin is excluded from training. This makes a narrow-origin artifact less plausible for general rural migration.

The climate result requires tighter wording. BEMP links recorded shocks to later moves, and BIHS directly records erosion-related land loss as a reason for relocation. Both samples show positive grouped performance. Neither small climate sample supports a general full-choice rule for an unseen origin. The

model's difficulty centers on whether the chosen alternative remains the origin district. Local land markets, damage intensity, relief, kin support, and the feasibility of rebuilding nearby may dominate that split.

Social networks deserve a separate mechanism study. Among the 1,857 BIHS interval migrants, 1,815 reported help from friends or family at the destination. That field is measured after destination selection and only for the chosen place. Entering it into candidate utility would leak the outcome because equivalent network measures do not exist for all 63 unchosen districts. A future survey should ask pre-move households where relatives, friends, employers, and prior migrants live, then score the same network measure for every candidate destination.

For planners, the result shifts attention from counts alone to receiving places. Districts differ in settlement, agricultural land, access to cities, and accumulated flood exposure. A destination model can help identify which places may receive movers under observed migration systems. It cannot substitute for housing, labor-market, infrastructure, or social-network data. Those measures should enter future work using the same candidate-wide and leakage-safe design.

8. Limitations

The public geography stops at districts. The analysis cannot distinguish a move across a river from a move across a district, recover neighborhood choice, or measure distance between exact homes. District averages also combine heterogeneous urban, rural, and hazard environments.

The samples measure different forms of mobility. BEMP includes whole- and partial-household relocation events. BIHS B4 records a household head's change of residence. BIHS V1 records current individual migrants away for at least six months. Agreement across these samples supports a predictive pattern, not a single migration process.

Timing limits causal interpretation. Some BIHS moves predate the GIS layers. BBS 2022 population postdates many events. The Global Flood Database is a selected-event archive rather than a complete flood climatology. Built surface, cropland, and travel time approximate destination structure but do not measure wages, rents, tenure, or actual travel.

Model validation also has finite support. The BEMP climate sample has seven observed origins. The BIHS erosion sample has 123 moves from 29 origins and only 71 cross-district events. Wide intervals in origin-held-out climate tests reflect that scarcity.

The public narrative preserves anonymity. Household identifiers, raw survey records, and event-level prediction tables should not be published in the open repository. The household illustration reports only district-level origin, destination, year, and the survey's categorical reason. No attempt was made to infer identity or fill gaps in the household's story.

9. Statements and declarations

Ethics approval and consent to participate

This study is a secondary analysis of anonymized public-use survey data. The author had no interaction with participants and no access to direct identifiers. The original BEMP data collection received approval from the ETH Zurich ethics committee (EK 2020-N-67), and participants provided informed consent (Freihardt 2025). BIHS received approval from the Institutional Review Board of the International Food Policy Research Institute (2014-44-PHND-D); the Bangladesh Ministry of Agriculture reviewed the survey instruments, and participants provided informed verbal consent. No additional participant consent was sought for this secondary analysis.

Data availability

BEMP public-use data are available through Zenodo at <https://doi.org/10.5281/zenodo.18229498>. BIHS public-use data are available through Harvard Dataverse for 2011-12 (<https://doi.org/10.7910/DVN/OR6MHT>), 2015 (<https://doi.org/10.7910/DVN/BXSYEL>), and 2018-19 (<https://doi.org/10.7910/DVN/NXKLZJ>). Analysis code, aggregate outputs, figures, and data-access instructions are available at <https://github.com/shailbelani23/bangladesh-climate-destination-choice>. The repository does not redistribute raw survey records, household identifiers, event ledgers, event-level predictions, or restricted derived flood layers. Researchers can reconstruct the analysis after obtaining the source data under their respective terms.

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Competing interests

The author declares no competing interests.

Author contributions

Shail Belani: conceptualization; data curation; formal analysis; investigation; methodology; software; visualization; writing—original draft; writing—review and editing.

10. Conclusion

An observed climate-related move contains two geographic facts: where the household left and where it arrived. This project models the second fact directly. Across BEMP and BIHS, destination flood history, built surface, urban accessibility, and cropland improve held-out district probabilities beyond population and distance.

The national migration model carries that gain across time and unseen origins. Climate-specific samples show that staying within an origin district depends more on local context. Research that joins pre-move

networks, housing, land access, jobs, and origin damage to candidate-wide destination data can test those mechanisms without weakening the revealed-choice design.

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Appendix

A. Event-ledger rules

A.1 BEMP

The prospective ledger begins with respondent-wave destination states. It excludes return observations that do not establish a new destination and the wave-6 migrant snapshot when no prior transition can be observed. Household linkage uses reconciled household keys rather than respondent identifiers alone. The climate indicator requires a flood or erosion record before the move interval. Ambiguous administrative labels are resolved through codebook text and a frozen district crosswalk. Events without a defensible district remain outside the model.

A.2 BIHS household relocations

Module B4 supplies the previous district, move year, move reason, and current Module A district. The erosion sample requires the codebook-verified response for loss of land or homestead land due to river erosion. The analysis excludes events lacking a valid origin, destination, or linkage key. The full household-relocation sample retains other reported reasons as a comparison, but reasons never enter candidate utility.

A.3 BIHS current migrants

The V1 ledger joins migration records to the household's current district and reconciles panel members across waves. The interval sample keeps 2015 and 2018-19 migrants whose current spell can be assigned to the survey interval without duplicating a migrant already observed at midline. International destinations and unresolved district names are excluded. All records from one household remain in the same validation fold.

B. GIS construction and provenance

District-level GIS processing follows one frozen 64-district boundary universe. Flood exposure unions valid non-permanent-water pixels from official Global Flood Database event rasters and intersects the union with a 2020 WorldCover land mask. Built surface uses the four 100-meter GHSL Mollweide tiles covering Bangladesh. Accessibility takes the pixelwise minimum across city-size classes at or above 50,000 people and reproduces the publisher's pre-combined layer. Cropland uses class 40 in the six WorldCover tiles intersecting Bangladesh.

Boundary cells receive fractional polygon intersections. Geographic rasters receive latitude-aware cell areas; GHSL remains in its equal-area projection. All 64 districts have complete values for the four final features. Source hashes, object versions, coverage measures, and extraction checks remain in the private reproducibility archive because the Global Flood Database license limits redistribution of derived layers.

C. Model and validation details

The direct GIS estimator minimizes conditional-logit negative log likelihood plus a ridge penalty on the four GIS coefficients. Distance and population coefficients remain unpenalized. Inner folds select the penalty. The final candidate grid is {0.001, 0.01, 0.1, 1, 10, 100}. The estimator uses stable log-sum-exp calculations and checks that candidate probabilities sum to one for every event.

The full validation suite contains 50 passing BIHS checks. It verifies 2,383 events with 64 alternatives and one chosen destination, zero household leakage, complete out-of-fold prediction keys, converged fits, unit probability sums, exact fold-origin matching, paired-estimate reproduction, and pre-specified transport warnings. Equivalent stage-specific checks cover BEMP event construction, choice sets, GIS extraction, model splits, convergence, and freeze hashes.

D. Interpretation guide

The supported estimand is destination choice conditional on an observed move. The full 64-district task also contains a within-origin alternative and therefore absorbs part of the stay-versus-cross decision. The interdistrict task conditions on crossing. Neither task estimates the causal effect of environmental shocks on migration incidence.

Log-loss gain is measured in natural-log units. A gain of 0.108 means that the chosen destination's geometric-mean probability is about 11.4% higher under GIS than gravity because $\exp(0.108) = 1.114$. This statement applies to average held-out probability, not to every household. Top-choice accuracy can remain unchanged when the model reallocates probability toward the observed destination without making it rank first, which is why log loss is the primary metric.

E. Public-release boundary

The public repository contains analysis code, aggregate result tables, validation summaries, figures, the manuscript, and data-access instructions. It excludes raw BEMP and BIHS files, event ledgers, household identifiers, event-level predictions, full choice sets, and restricted derived flood layers. Researchers can reconstruct those files after obtaining the source surveys under their respective terms.